

Chapter 4 Elementary Number Theory

4.1 Real Numbers

Real numbers (denoted as $\mathbf{R}$) consists of integers ($\mathbf{Z}$), rational numbers ($\mathbf{Q}$) and irrational numbers.

Definition

Let n be an integer.

n is even $\Leftrightarrow$ there exist an integer k such that $n = 2k$.

n is odd $\Leftrightarrow \exists$ an integer k such that $n = 2k + 1$.

Example 4.11

Use the definition to justify your answers to the following questions.

- (a) Is 0 even ?
- (b) Is -301 odd?
- (c) If a and b are integers, is $6a^2b$ even ?
- (d) If a and b are integers, is $10a + 8b + 1$ odd?

Solution :

- (a) Yes, $0 = 2 \cdot 0$.
- (b) Yes, $-301 = 2(-151) + 1$
- (c) Yes, $6a^2b = 2(3a^2b)$, since a and b are integers, so is $3a^2b$.
- (d) Yes, $10a + 8b + 1 = 2(5a + 4b) + 1$, since a and b are integers, so is $5a + 4b$.

Theorem 4.11

- (1) The sum of two even numbers is even
- (2) The sum of two odd numbers is even
- (3) The sum of an even number and an odd number is odd.

Proof :

- (1) Suppose m and n are even integers.

By definition of even, $m = 2r$, $n = 2s$ for some integers r and s .

$$\begin{aligned} \text{Then } m + n &= 2r + 2s \\ &= 2(r + s) \\ &= 2k \quad (\text{let } k = r + s, k \text{ is an integer}) \end{aligned}$$

$\therefore$ By definition of even, $m + n$ is even.

Definition

Let n be an integer and $n > 1$. Suppose $n = r \cdot s$ for any positive integers r and s .
 n is prime if and only if $r = 1$ or $s = 1$.
 If n is not prime, n is composite.

Example 4.1.2

- (a) Is 1 prime?
- (b) Write the first 6 prime numbers.
- (c) Write the first 6 composite numbers.

Solution :

- (a) No, a prime number is required to be greater than 1.
- (b) 2, 3, 5, 7, 11, 13
- (c) 4, 6, 8, 9, 10, 12

Definition

A real number r is rational if and only if it can be expressed as a quotient of two integers with a nonzero denominator. A real number that is not rational is irrational number.

Let $r \in \mathbb{R}$. r is rational $\Leftrightarrow \exists$ integers a and b such that $r = \frac{a}{b}$ and $b \neq 0$.

Example 4.1.3

- (a) Is $\frac{10}{3}$ a rational number ?
- (b) Is $-\frac{5}{39}$ a rational number ?
- (c) Is 7 a rational number?
- (d) Is 0 a rational number ?
- (e) Is $\frac{2}{0}$ a rational number ?
- (f) Is $\frac{2}{0}$ an irrational number ?

Solution :

- (a) Yes, $\frac{10}{3}$ is a quotient of the integers 10 and 3.
- (b) Yes, $-\frac{5}{39}$ is a quotient of the integers -5 and 39.
- (c) Yes, $7 = \left(\frac{7}{1}\right)$.
- (d) Yes, $0 = \left(\frac{0}{1}\right)$.
- (e) No, $\frac{2}{0}$ is not a number (division by 0 is not allowed).
- (f) No, because every irrational number is a number, and $\frac{2}{0}$ is not a number.

Theorem 4.1.2

- (1) Every integer is a rational number.
- (2) Sum of two rational number is rational.

Proof : (2)

Suppose r and s are rational number. Then, by definition of rational, $r = \frac{a}{b}$, $s = \frac{c}{d}$ for some integer a, b, c and d with $b \neq 0$, $d \neq 0$.

$$\begin{aligned}
 r + s &= \frac{a}{b} + \frac{c}{d} \\
 &= \frac{ad + bc}{bd}, \quad \text{let } p = ad + bc, q = bd
 \end{aligned}$$

$$= \frac{p}{q}, \text{ where } p \text{ and } q \text{ are integer, } q \neq 0.$$

$\therefore r + s$ is rational by definition of a rational number.

4.2 Divisibility

Definition

If n and d are integers, then n is divisible by d if and only if $n = dk$ for some integer k .

The notation $d|n$ is read “ d divides n ”.

Remark : $d \nmid n \Leftrightarrow \frac{n}{d} \notin \mathbb{Z}$.

Theorem 4.2.1 For all integers a , b and c , if $a|b$ and $b|c$, then $a|c$.

Proof :

Suppose a , b and c are integers such that a divides b and b divides c .

By definition of divisibility ,

$$b = ar \quad \text{and } c = bs \text{ for some integers } r \text{ and } s.$$

$$\begin{aligned} \text{By substitution} \quad c &= bs \\ &= (ar)s \\ &= a(rs) \\ &= ak \quad (\text{let } k = rs) \end{aligned}$$

$\therefore a$ divides c by definition of divisibility.

Example 4.2.1: Does $4|15$?

Solution :

$$\text{No, } \frac{15}{4} = 3.75, \text{ which is not an integer.}$$

Theorem 4.2.2 (Unique Factorization Theorem for the Integers)

Given any integer $n > 1$, there exist a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k} \quad (\text{standard factor form})$$

and any other expression of n as a product of prime numbers is identical to this except, perhaps, for the order in which the factors are written.

Example 4.2.2

Write 3300 in standard factored form.

Solution :

$$\begin{aligned} 3300 &= 100 \times 33 = 4 \times 25 \times 3 \times 11 \\ &= 2 \times 2 \times 5 \times 5 \times 3 \times 11 \\ &= 2^2 \cdot 3^1 \cdot 5^2 \cdot 11^1 \end{aligned}$$

4.3 Quotient – Remainder Theorem

Example 4.3.1: Divide 11 by 4

Solution :

$$\begin{array}{r} 2 \\ 4 \overline{)11} \\ \underline{8} \\ 3 \end{array} \quad \therefore 11 = 2 \cdot 4 + 3$$

Theorem 4.3.1 (Quotient- Remainder Theorem)

Given any integer n and positive integer d there exist unique integers q and r such that

$$n = qd + r \quad \text{and} \quad 0 \leq r < d.$$

Example 4.3.2

For each values of n and d , find integers q and r such that $n = qd + r$ and $0 \leq r < d$.

- (a) $n = 54, d = 4$
- (b) $n = -54, d = 4$
- (c) $n = 54, d = 70$

Solution :

- (a) $54 = 4 \cdot 13 + 2$; $q = 13, r = 2$
- (b) $-54 = 4 \cdot (-14) + 2$; $q = -14, r = 2$
- (c) $54 = 70 \cdot 0 + 54$; $q = 0, r = 54$

Definition

If n and d positive integers, then $n(\bmod d) = r \Leftrightarrow n = qd + r$ where q and r integers and $0 \leq r < d$.

Example 4.3.3

- (a) Compute $32 (\bmod 9)$?
- (b) What day of the week it be 1 year from today ?

Solution :

$$\begin{array}{r} 3 \\ 9 \overline{) 32} \\ \underline{27} \\ 5 \end{array}$$

$$\begin{aligned} 32 (\bmod 9) &= 32 \bmod 9 = 5 \\ 32 (\text{div } 9) &= 3 \end{aligned}$$

- (b) Suppose today is Tuesday and there are 365 days in a year that is not a leap year, and each week has 7 days.

$$\begin{aligned} 365 &= 7 \cdot 52 + 1 & 365 \text{ div } 7 &= 52 \\ & & 365 \bmod 7 &= 1 \end{aligned}$$

Thus 52 week (364 days), from today will be a Tuesday, so 365 days from today will be 1 day later, namely Wednesday.

$$365 (\bmod 7) = 1$$

** If there are 366 days in a leap year, then $366 = 7 \times 52 + 2$ (2 days later from today)

Definition

Let $x \in \mathbb{R}$ and $n \in \mathbb{Z}$.

- (1) The floor of x , denoted as $\lfloor x \rfloor$, is defined as $\lfloor x \rfloor = n$ where $n \leq x < n + 1$.
- (2) The ceiling of x , denoted as $\lceil x \rceil$, is defined as $\lceil x \rceil = n$ where $n - 1 < x \leq n$.

Example 4.3.4

Compute $\lfloor x \rfloor$ and $\lceil x \rceil$ for each of the following values of x .

- (a) $\frac{25}{4}$ (b) 0.999 (c) -2.01

Solution :

$$(a) \quad \frac{25}{4} = 6.25 \quad \Rightarrow \quad \left\lfloor \frac{25}{4} \right\rfloor = 6$$

$$\Rightarrow \quad \left\lceil \frac{25}{4} \right\rceil = 7$$

$$(b) \quad \begin{aligned} \lfloor 0.999 \rfloor &= 0 \\ \lceil 0.999 \rceil &= 1 \end{aligned}$$

$$(c) \quad \begin{aligned} \lfloor -2.01 \rfloor &= -3 \\ \lceil -2.01 \rceil &= -2 \end{aligned}$$

Example 4.3.5

The 1370 soldiers at a military base are given the opportunity to take buses into town for an evening out. Each bus holds a maximum of 40 passengers.

- (a) For reasons of economy, the base commander will send only full buses. What is the maximum number of buses the base commander will send ?
- (b) If the base commander is willing to send a partially filled bus, how many buses will the commander need to allow all the soldiers to take the trip ?

Solution :

$$(a) \quad \left\lfloor \frac{1370}{40} \right\rfloor = \lfloor 34.25 \rfloor = 34$$

$$(b) \quad \left\lceil \frac{1370}{40} \right\rceil = \lceil 34.25 \rceil = 35$$

Example 4.3.6

If k is an integer, what are $\lfloor k \rfloor$ and $\left\lfloor k + \frac{1}{2} \right\rfloor$? why?

Solution :

Suppose k is an integer ,

Then $\lfloor k \rfloor = k$ because k is an integer and $k \leq k < k+1$.
 $\left\lfloor k + \frac{1}{2} \right\rfloor = k$ because k is an integer and $k \leq k + \frac{1}{2} < k+1$.

Example 4.3.7

Is $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$? why?

Solution :

No, The statement is false. As a counterexample, take $x = y = \frac{1}{2}$. Then,

$$\lfloor x \rfloor + \lfloor y \rfloor = \left\lfloor \frac{1}{2} \right\rfloor + \left\lfloor \frac{1}{2} \right\rfloor = 0 + 0 = 0$$

$$\lfloor x + y \rfloor = \left\lfloor \frac{1}{2} + \frac{1}{2} \right\rfloor = \lfloor 1 \rfloor = 1$$

$$\therefore \lfloor x + y \rfloor \neq \lfloor x \rfloor + \lfloor y \rfloor$$

Theorem 4.3.2

For all real numbers x and all integers m , $\lfloor x + m \rfloor = \lfloor x \rfloor + m$.

Proof:

Suppose a real number x and an integer m are given .

Let $n = \lfloor x \rfloor$. By definition of floor, n is an integer and $n \leq x < n+1$

Add m to all sides to obtain

$$n + m \leq x + m < n + m + 1$$

Now, $n + m$ is an integer (since n and m are integers and a sum of integers is an integer).

By definition of floor,

$$\begin{aligned} \lfloor x + m \rfloor &= n + m \\ &= \lfloor x \rfloor + m \quad (\because n = \lfloor x \rfloor) \end{aligned}$$

Theorem 4.3.3

For any integer n ,

$$\left\lfloor \frac{n}{2} \right\rfloor = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ \frac{n-1}{2} & \text{if } n \text{ is odd} \end{cases}$$

Proof :

Suppose n is an integer. By the quotient-remainder theorem, n is odd or n is even.

Case 1 (n is odd) : In this case, $n = 2k + 1$ for some integer k . The left-hand side of the equation

$$\left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{2k+1}{2} \right\rfloor = \left\lfloor k + \frac{1}{2} \right\rfloor = k \quad (\because k \leq k + \frac{1}{2} < k+1)$$

And the right-hand side of the equation is

$$\frac{n-1}{2} = \frac{2k+1-1}{2} = k$$

Since both the left-hand and right-hand sides equal k ,

$$\therefore \left\lfloor \frac{n}{2} \right\rfloor = \frac{n-1}{2} \quad (\text{if } n \text{ is odd})$$

Case 2 (n is even) : In this case, $n = 2k$ for some integer k .
The left-hand side of the equation is

$$\left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{2k}{2} \right\rfloor = \lfloor k \rfloor = k \quad (\because k \leq k < k+1)$$

and the right-hand side of the equation is

$$\frac{n}{2} = \frac{2k}{2} = k$$

Since both the left-hand and right-hand sides equal k ,

$$\therefore \left\lfloor \frac{n}{2} \right\rfloor = \frac{n}{2} \quad (\text{if } n \text{ is even}).$$

4.4 Euclidean algorithm

Definition

Let a and b be integers that are not both zero. The greatest common divisor of a and b , denoted $\gcd(a, b)$, is that integer d with the following properties :

1. d is a common divisor of both a and b . In other words, $d|a$ and $d|b$.
2. For all integers c , if c is a common divisor of both a and b , then c is less than or equal to d . In other words,

for all integers c , if $c|a$ and $c|b$, then $c \leq d$.

Example 4.4.1

- (a) Find $\gcd(72, 63)$
- (b) Find $\gcd(10^{20}, 6^{30})$.

Solution :

- (a) $72 = 9 \cdot 8$ so $9|72$, $9|63$, and no integer larger than 9 divides both 72 and 63.

$$63 = 9 \cdot 7$$

$$\therefore \gcd(72, 63) = 9$$

- (b) By the laws of exponents, $10^{20} = 2^{20} \cdot 5^{20}$
 $6^{30} = 2^{30} \cdot 3^{30} = 2^{20} \cdot 2^{10} \cdot 3^{30}$

It follows that $2^{20}|10^{20}$, $2^{20}|6^{30}$,

$$\therefore \gcd(10^{20}, 6^{30}) = 2^{20}.$$

The Euclidean Algorithm:

Put $r_0 = m$ and $r_1 = a$. We perform the following sequence of divisions with remainder:

$$r_0 = q_1 r_1 + r_2, \quad 0 < r_2 < r_1$$

$$r_1 = q_2 r_2 + r_3, \quad 0 < r_3 < r_2$$

⋮

$$r_{n-2} = q_{n-1} r_{n-1} + r_n, \quad 0 < r_n < r_{n-1}$$

$$r_{n-1} = q_n r_n$$

Then we have $\gcd(m, a) = r_n$. In other words, $\gcd(m, a)$ is the last nonzero remainder in the above divisions.

Example 4.4.2

(a) Find $\gcd(330, 156)$

Solution :

$$330 = 156 \cdot 2 + 18$$

$$156 = 18 \cdot 8 + 12$$

$$18 = 12 \cdot 1 + 6$$

$$12 = 6 \cdot 2 + 0$$

$$\begin{aligned} \therefore \gcd(330, 156) &= \gcd(156, 18) \\ &= \gcd(18, 12) \\ &= \gcd(12, 6) \\ &= \gcd(6, 0) \\ &= 6 \end{aligned}$$

(b)

Find $\gcd(662, 414)$

Solution :

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41 + 0$$

$$\therefore \gcd(662, 414) = 2$$

Chinese Remainder Theorem

Two integers a and b are said to be relatively prime (or coprime) if $\gcd(a, b) = 1$.

If $\alpha a = 1 \pmod{m}$, we say that α is the inverse of a modulo m and is denoted by

$$\alpha = a^{-1} \pmod{m}$$

$a^{-1} \pmod{m}$ exists if and only if $\gcd(a, m) = 1$.

Example :

Find an inverse of 3 modulo 7.

Solution :

Since $\gcd(3, 7) = 1$, an inverse of 3 modulo 7 exists:

$$7 = 2 \cdot 3 + 1$$

$$-2 \cdot 3 + 1 \cdot 7 = 1$$

$\therefore -2$ is an inverse of 3 modulo 7.

(or $-2+7=5$ is a positive inverse of 3 modulo 7)

The Extended Euclidean Algorithm

Suppose $\gcd(m, a) = 1$ and let $q_1, q_2, \dots$ be the quotients in Euclidean algorithm.

Put $t_0 = 0, t_1 = 1$ and

$$t_j = t_{j-2} - q_{j-1}t_{j-1} \pmod{m} \text{ for } j = 2, \dots, n.$$

Then $t_n = a^{-1} \pmod{m}$.

Example :

Determine whether $a^{-1} \pmod{m}$ exists. If it exists, find $a^{-1} \pmod{m}$. If it does not exist, explain.

$$112^{-1} \pmod{543}$$

Solution :

$$\begin{aligned} t_2 &= 0 - 4(1) = -4 \\ t_3 &= 1 - 2(-4) = 9 \\ t_4 &= -4 - 19(9) = -175 \\ t_5 &= 9 - 1(-175) = 184 \end{aligned}$$

$$\therefore 121^{-1} \pmod{543} = 184$$

Chinese Remainder Theorem

Let $m_1, m_2, \dots, m_r$ be pairwise coprime positive integers and let $a_1, a_2, \dots, a_r$ be arbitrary integers. Then there exists an integer x with $x = a_i \pmod{m_i}$ for $1 \leq i \leq r$ and x is uniquely determined modulo $M = m_1 m_2 \dots m_r$ which is given by

$$x = \sum_{i=1}^r a_i M_i y_i \pmod{M}$$

Where $M_i = \frac{M}{m_i}$ and $y_i = M_i^{-1} \pmod{m_i}$ for $1 \leq i \leq r$.

Use Chinese remainder theorem to compute :

(a) $x \pmod{35}$, if $x = 3 \pmod{5}$, $x = 6 \pmod{7}$.

(b) $x \pmod{210}$, if $x = 1 \pmod{2}$, $x = 1 \pmod{3}$, $x = 3 \pmod{7}$ and $x = 3 \pmod{5}$.

Solution :

$$(a) \quad a_1 = 3, \quad a_2 = 6$$

$$m_1 = 5, \quad m_2 = 7$$

$$M_1 = 7, \quad M_2 = 5$$

$$\begin{aligned} y_1 &= 7^{-1} \pmod{5}, & y_2 &= 5^{-1} \pmod{7} \\ &= 2^{-1} \pmod{5} & &= 3 \\ &= 3 & & \end{aligned}$$

$$\begin{aligned} x &= \sum a_i M_i Y_i \pmod{M} \\ &= 3(7)(3) + 6(5)(3) \end{aligned}$$

$$\begin{aligned} \therefore &= 63 + 90 \\ &= 153 \pmod{35} \\ &= 13 \pmod{35} \end{aligned}$$

$$(b) \quad \begin{array}{llll} a_1 = 1, & a_2 = 1, & a_3 = 3, & a_4 = 3 \\ m_1 = 2, & m_2 = 3, & m_3 = 7, & m_4 = 5 \end{array}$$

$$M_1 = 105, \quad M_2 = 70, \quad M_3 = 30, \quad M_4 = 42$$

$$\begin{array}{llll} y_1 = 105^{-1} \pmod{2} & y_2 = 70^{-1} \pmod{3} & y_3 = 30^{-1} \pmod{7} & y_4 = 42^{-1} \pmod{5} \\ = 1^{-1} \pmod{2} & = 1^{-1} \pmod{3} & = 2^{-1} \pmod{7} & = 2^{-1} \pmod{5} \\ = 1 & = 1 & = -3 & = 3 \\ & & = 4 & \end{array}$$

$$\begin{aligned} \therefore x &= 1(105)(1) + 1(70)(1) + 3(30)(4) + 3(42)(3) \\ &= 913 \pmod{210} \\ &= 73 \pmod{210} \end{aligned}$$

4.5 Mathematical Induction

Mathematical induction is one of the more recently developed techniques of proof in the history of mathematics. It is used to check conjectures about the outcomes of processes that occur repeatedly and according to definite patterns.

Principle of Ordinary Mathematical Induction

Let $P(n)$ be a predicate that is defined for integers n , and let a be a fixed integer. Suppose the following two statements are true:

- (1) $P(a)$ is true.
- (2) For all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true.

Then the statement “for all integers $n \geq a$, $P(n)$ ” is true.

Remark:

Proving a statement by mathematical induction is a two-step process.

In step 1, the **basis step**, we prove that $P(a)$ is true for a particular integer a .

In step 2, the **inductive step**, we prove that for all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true.

To prove the inductive step, we suppose that $P(k)$ is true, where k is a particular but arbitrarily chosen integer greater than or equal to a . Then we show that $P(k+1)$ is true. The supposition that $P(k)$ is true is called the **inductive hypothesis**.

Note:

In a proof by mathematical induction, it is not assumed that $P(k)$ is true for all positive integers!

Examples Of Proofs By Mathematical Induction

Example 4.5.1:

Use mathematical induction to prove that the sum of the first n odd positive integers is n^2 .

Solution:

$P(n)$: The sum of the first n odd positive integers is n^2 .

Basis step:

$P(1)$ is true since the sum of the first odd positive integer is 1.

Inductive step:

Suppose that $P(k)$ is true for a positive integer $k \geq 1$; that is,

$$1 + 3 + 5 + \dots + (2k - 1) = k^2.$$

We must show that $P(k + 1)$ is true.

But,

$$\begin{aligned} 1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) &= k^2 + (2k + 1) \\ &= (k + 1)^2. \end{aligned}$$

Hence $P(k + 1)$ is true.

Since $P(1)$ is true and the implication $P(k) \rightarrow P(k + 1)$ is true for all positive integers k , the principle of mathematical inductions shows that $P(n)$ is true for all positive integers n .

Example 4.5.2: (Sum of the first n integers)

Use mathematical induction to prove that $\sum_{r=1}^n r = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$

Proof:

The formula is true for $n = 1$ because $\frac{1(1+1)}{2} = \frac{2}{2} = 1$.

Suppose $1 + 2 + \dots + k = \frac{k(k+1)}{2}$, for some integer $k \geq 1$.

$$1 + 2 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Since we have proved both the basis and inductive steps, we conclude that the formula is true.

Remark: 1 2 3 50 51 98 99 100

Example 4.5.3:

(i) Use mathematical induction to prove that

$$\sum_{r=1}^n r^2 = 1^2 + 2^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1).$$

Solution

$$P(n): \sum_{r=1}^n r^2 = 1^2 + 2^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$$

Basis step:

$$n = 1, \text{ LHS} = 1 \text{ and } \text{RHS} = \frac{1}{6}(2)(3) = 1$$

So LHS = RHS and $P(1)$ is true

Inductive step:

Suppose that $P(k)$ is true for a positive integer $k \geq 1$; that is,

$$1^2 + 2^2 + \dots + k^2 = \frac{1}{6}k(k+1)(2k+1)$$

When $n = k + 1$

$$\begin{aligned} (1^2 + 2^2 + \dots + k^2) + (k+1)^2 &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\ &= \frac{1}{6}(k+1)[2k^2 + k + 6k + 6] \\ &= \frac{1}{6}(k+1)(k+2)(2k+3) \end{aligned}$$

Hence $P(k+1)$ is true.

Since $P(1)$ is true and the implication $P(k) \rightarrow P(k+1)$ is true for all positive integers k , the principle of mathematical inductions shows that $P(n)$ is true for all positive integers n .

Example 4.5.4: (The Tower of Hanoi)

Let A, B and C be 3 poles with $n \geq 1$ discs of different diameters placed on pole A such that the largest is on the bottom and the smallest on top. Move the entire pile of discs from pole A to pole C by moving one disc at a time and never placing a disc on top of a smaller one. Prove that the minimum number of moves required is $2^n - 1$.

Proof:

The proof is by induction on n .

Basis step:

If $n = 1$, clearly the minimum number of moves required is $1 = 2^1 - 1$.

Inductive step:

Suppose the minimum number of moves required for k discs is $2^k - 1$ for some $k \geq 1$.

Let there be $k + 1$ discs on pole A. Then we have the following:

the minimum number of moves required to move the top k discs from pole A to pole B is $2^k - 1$.

the minimum number of moves required to move the one last (largest) disc from pole A to pole C is 1.

the minimum number of moves required to move all the k discs from pole B to pole C is $2^k - 1$.

Hence the minimum number of moves required to move all the $k + 1$ discs from pole A to pole C is

$$(2^k - 1) + 1 + (2^k - 1) = 2^{k+1} - 1.$$

Therefore the minimum number of moves required to move all the n discs from pole A to pole C is $2^n - 1$ for any $n \geq 1$. This completes the proof.

Example 4.5.6: (Proving a Divisibility Property)

- (i) Use mathematical induction to prove that for all integers $n \geq 1$, $2^{2n} - 1$ is divisible by 3.

Solution:

$P(n)$: $2^{2n} - 1$ is divisible by 3.

$P(1)$ is true because $2^{2(1)} - 1$ is divisible by 3.

Suppose $2^{2k} - 1$ is divisible by 3 for some integer $k \geq 1$.

We must show that $2^{2(k+1)} - 1$ is divisible by 3.

But $2^{2(k+1)} - 1 = 2^{2k+2} - 1 = 2^{2k}(3) + (2^{2k} - 1)$.

By the inductive hypothesis, $2^{2k} - 1$ is divisible by 3 and clearly $2^{2k}(3)$ is divisible by 3. Hence, by induction, $2^{2n} - 1$ is divisible by 3 for all integers $n \geq 1$.

- (ii) Use mathematical induction to prove that for all integers $n \geq 1$, $7^n(3n + 1) - 1$ is divisible by 9.

$P(1)$ is true because $7(3 + 1) - 1 = 27$ is divisible by 9.

Suppose $7^k(3k + 1) - 1$ is divisible by 9 for some integer $k \geq 1$.

We must show that $7^{k+1}(3(k + 1) + 1) - 1$ is divisible by 9.

$$\begin{aligned} \text{But } 7^{k+1}(3(k + 1) + 1) - 1 &= 7(7^k)[3k + 4] - 1 \\ &= 7^k(21k + 28) - 1 \end{aligned}$$

$$\begin{aligned}
 &= \left[7^k(3k+1) - 1 \right] + 7^k(18k+27) \\
 &= \left[7^k(3k+1) - 1 \right] + (9)7^k(2k+3)
 \end{aligned}$$

By the inductive hypothesis, $7^k(3k+1) - 1$ is divisible by 9 and clearly $(9)7^k(2k+3)$ is divisible by 9.

Hence, by induction, $7^n(3n+1) - 1$ is divisible by 9 for all integers $n \geq 1$.

4.6 Principle of Strong Mathematical Induction

Let $P(n)$ be a predicate that is defined for integers n , and let a, b be fixed integers with $a \leq b$.

Suppose the following two statements are true:

- (1) $P(a), P(a+1), \dots$, and $P(b)$ are all true
- (2) For any integer $k > b$, if $P(i)$ is true for all integers i with $a \leq i < k$, then $P(k)$ is true.

Then the statement “for all integers $n \geq a$, $P(n)$ ” is true.

Remark:

Statement (1) is called the basis step.

Statement (2) is called the inductive step.

Example 4. 20:

- (i) Prove that any integer greater than 1 is divisible by a prime number.

Solution:

$P(n)$: n is divisible by a prime number, $n \geq 2$.

Basis step:

$P(2)$ is true because 2 is a prime number and 2 is divisible by itself.

Inductive step:

(If the divisibility property holds for all i with $2 \leq i < k$, then it holds for k)

Let $k > 2$.

Suppose that for all integers i with $2 \leq i < k$, i is divisible by a prime number.

Case 1:

k is prime.

Then k is divisible by a prime number, namely itself.

Case 2:

k is not a prime.

Then $k = ab$ where $2 \leq a < k$ and $2 \leq b < k$ where a, b are integers.

By the inductive hypothesis, a is divisible by a prime number p .

Hence k is divisible by p .

By strong mathematical induction, $P(n)$ is true.

- (ii) Show that if n is an integer greater than 1, then n can be written as the product of primes.

Solution:

$P(n)$: n can be written as the product of primes

Basis step:

$P(2)$ is true because 2 can be written as the product of one prime, namely itself.

Inductive step:

(If the divisibility property holds for all i with $2 \leq i < k$, then it holds for k)

Let $k > 2$.

Suppose that for all integers i with $2 \leq i < k$, i can be written as the product of primes.

Case 1:

k is prime.

Then k can be written as the product of one prime, namely itself.

Case 2:

k is not a prime.

Then $k = ab$ where $2 \leq a < k$ and $2 \leq b < k$ where a, b are integers.

By the inductive hypothesis, a and b can be written as the product of primes.

Hence k can be written as the product of primes.

By strong mathematical induction, $P(n)$ is true.

- (iii) Define a sequence $a_1, a_2, a_3, \dots$ as follows:

$$a_1 = 0$$

$$a_2 = 2$$

$$a_k = 3a_{\lfloor k/2 \rfloor} + 2, \quad k \geq 3.$$

Find the first 7 terms of the sequence and prove that a_n is even for each integer $n \geq 1$.

Solution:

$$a_1 = 0$$

$$a_2 = 2$$

$$a_3 = 3a_{\lfloor 3/2 \rfloor} + 2 = 3a_1 + 2 = 2$$

$$a_4 = 3a_{\lfloor 4/2 \rfloor} + 2 = 3a_2 + 2 = 8$$

$$a_5 = 3a_{\lfloor 5/2 \rfloor} + 2 = 3a_2 + 2 = 8$$

$$a_6 = 3a_{\lfloor 6/2 \rfloor} + 2 = 3a_3 + 2 = 8$$

$$a_7 = 3a_{\lfloor 7/2 \rfloor} + 2 = 3a_3 + 2 = 8$$

$P(n)$: a_n is even

Basis step:

$P(n)$ holds for $n = 1$ and $n = 2$ since $a_1 = 0$ and $a_2 = 2$ are even.

Inductive step:

(If the divisibility property holds for all i with $1 \leq i < k$, then it holds for k)

Let $k > 2$.

Suppose that a_i is even for all integers i with $1 \leq i < k$.

By definition of $a_1, a_2, a_3, \dots$

$$a_k = 3a_{\lfloor k/2 \rfloor} + 2, \text{ for all integers } k \geq 3.$$

Now $a_{\lfloor k/2 \rfloor}$ is even by inductive hypothesis {because $k > 2$ and so $1 \leq \lfloor k/2 \rfloor < k$ }. Thus

$3a_{\lfloor k/2 \rfloor}$ is even {because odd $\times$ even = even}, and hence $3a_{\lfloor k/2 \rfloor} + 2$ is even.

Consequently, a_k , which equals $3a_{\lfloor k/2 \rfloor} + 2$, is even.

By strong mathematical induction, $P(n)$ is true.

4.7 Well-Ordering Principle for Integers

Let S be a set containing one or more integers all of which are greater than some fixed integers.

Then S has a least element.

Example 4.21: (Finding least elements)

In each case, if the set has a least element, state what it is. If not, explain why the well-ordering principle is not violated.

- (i) The set of all positive real numbers

There is no least positive real number. For if x is any positive real number, then $x/2$ is a positive real number that is less than x . No violation of the well-ordering principle occurs because the well-ordering principle refers only to sets of integers and this set is not a set of integers.

- (ii) The set of all nonnegative integers n such that $n^2 < n$.

There is no least nonnegative integer n such that $n^2 < n$ because there is no nonnegative integer that satisfies this inequality. The well-ordering principle is not violated because the well-ordering principle refers to sets that contain at least one or more elements.

- (iii) The set of all nonnegative integers of the form $46 - 7k$, where k is an integer.

The following table shows values of $46 - 7k$ for various values of k .

k	...	-3	-2	-1	0	1	2	3	4	5	6	7	...
$46 - 7k$	...	67	60	53	46	39	32	25	18	11	4	-3	...

The table suggests, and you can easily confirm, that $46 - 7k < 0$ for $k \geq 7$ and that $46 - 7k \geq 46$ for $k \leq 0$. There, from the other values in the table it is clear that 4 is the **least nonnegative** integer of the form $46 - 7k$. This corresponds to $k = 6$.

Division Algorithm:

If a is an integer and d is a positive integer, then there are unique integers q and r with

$$0 \leq r < d \text{ and } a = dq + r.$$

Example 4.22:

Use the well-ordering principle to prove the division algorithm.

Proof :

Quotient-Remainder Theorem

Let S be the set of all nonnegative integers of the form

$$n - d \cdot k$$

where k is an integer. This set has at least one element.

$$[n \text{ is nonnegative} \Rightarrow n - 0 \cdot d = n \geq 0 \\ \text{and so } n - 0 \cdot d \text{ is in } S.]$$

$$n \text{ is negative} \quad \Rightarrow \quad n - n \cdot d = n(1 - d) \geq 0 \\ \text{and so } n - n \cdot d \text{ is in } S.]$$

It follows by the well-ordering principle that S contains a least element r . Then for some specific integer $k = q$,

$$n - d \cdot q = r$$

[because every integer in S can be written in this form].

$$\text{So } n = d \cdot q + r$$

Furthermore, $r < d$.

[suppose $r \geq d$. Then

$$n - d \cdot (q + 1) = n - d \cdot q - d = r - d \geq 0$$

and so $n - d \cdot (q + 1)$ would be a nonnegative integer in S that would be smaller than r .

But r is the smallest integer in S . This contradiction shows that the supposition $r \geq d$ must be false]

The preceding arguments prove that there exist integers r and q for which

$$n = d \cdot q + r \quad \text{and} \quad 0 \leq r < d$$

4.8 other methods of proving

Example 4.8.1

Disprove the following by giving a counterexample.

For all real numbers a and b , if $a^2 = b^2$ then $a = b$.

Solution :

$$\text{Let } a = 1 \text{ and } b = -1$$

$$\begin{aligned} \text{Then, } a^2 &= 1^2 = 1 \\ b^2 &= (-1)^2 = 1 \end{aligned}$$

$$\therefore a^2 = b^2$$

$$\text{But } a \neq b \quad \text{since} \quad 1 \neq -1.$$

Theorem 4.8.1

Suppose a and b are positive integers and $a|b$. Then $a \leq b$.

Proof :

$a|b$ means that $b = ka$ for some integer k .

k must be a positive integer because $a > 0$, $b > 0$.

$$1 \leq k$$

$$a \cdot 1 \leq k \cdot a$$

Theorem 4.8.2 There is no greatest integer .

Proof :

[Using negation of the theorem and suppose it to be true].

Suppose there is a greatest integer N .

Then $N \geq n$ for every integer n .

Let $M = N + 1$. Now M is an integer since it's a sum of integers.

$$\text{Also } M > N \text{ (} \because M = N + 1 \text{)}$$

Thus, M is an integer that is greater than N ,

So, N is the greatest integer and N is not the greatest integer, which is a contradiction.
Hence, theorem is true.

Theorem 4.8.3

There is no integer that is both even and odd.

Proof :

Suppose there is an integer n that is both even and odd.

By definition of even, $n = 2a$ for some integer a , and

By definition of odd, $n = 2b + 1$. for some integer b ,

Consequently,

$$2a = 2b + 1$$

$$2a - 2b = 1$$

$$2(a - b) = 1$$

$$(a - b) = \frac{1}{2}$$

Since a and b are integers, the difference $a - b$ must also be an integer, but $a - b$ is not an integer. Thus, $a - b$ is an integer and $a - b$ is not an integer which is a contradiction.

$\therefore$ The theorem is true.

Theorem 4.8.4 . For all integers, if n^2 is even then n is even.

Proof :

Suppose n is any odd integer.

By definition of odd, $n = 2k + 1$ for some integer k .

By substitution and algebra,

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

But $2k^2 + 2k$ is an integer because products and sums of integers are integers.

So, $n^2 = 2(\text{an integer}) + 1$.

Thus by def. of odd, n^2 is odd.

Contrapositive (“ n is odd, the n^2 is odd”) is also true.

$\therefore$ If n^2 is not odd, then n is not odd.
